

Standards,
Precautions &
Advances in
Ancient
Metagenomics

SPAAM Newsletter #8

Hello from the SPAAM Steering Committee in the New Year 2026!

As winter settles in and the world is in its seasonal quiet, the SPAAM community is anything but dormant! We're moving full speed into 2026, and we're excited to bring you the latest updates, new articles, upcoming events, and exciting opportunities. 🧬 🦴

We hope you enjoy this edition of the SPAAM Newsletter!

Aida, Marcel, Maria, Mariana, Joanna, Davide, Rémi, Laura, Sierra, Wenqin and Anastasia

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Announcements

SPAAM Constitution

The Steering Committee is happy to announce that the new constitution has been formally accepted through a ballot ending on Dec 19, with 13/13 YES votes. The steering committee will now move forward to implement the new constitution and plan the first election for the Steering Committee in summer 2026.

Hackathon for AMDir

The AncientMetagenomeDir is having its next metadatathon on the 6th of February!

AncientMetagenomeDir is a community project that aims to collect all the metadata for all the published ancient metagenomic datasets. If you want to learn more about the project, please visit our [website](#) and our [Github page](#).

We have made the experience that metadatathon are the most efficient way to keep the database updated, so we want to take this occasion to invite you to the next one and to share some information about our event on February 6th! For this metadatathon we have the following objectives: compile radiocarbon dates or archaeological/historical dates for the host associated ancient metagenomes, tackle any open issues in AncientMetagenomeDir, review documentation, update amdirt, the accompanying toolkit for AncientMetagenomeDir.

We will provide initial training on how to introduce data to the GitHub repository, as well as on interpretation of radiocarbon

dates, and we will work together to collect all the information for the C14 dates, close any open issues in AMDir and update our docs and tooling. Join us in this effort and make your future self happy!

When?

6th of February 2026, and we will adjust the actual hours to the time zones of the participants. So please feel welcome from any part of the world!

Where?

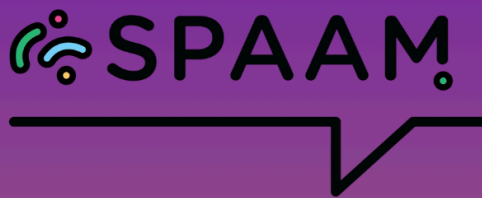
virtual, link and platform details will be sent before the event.

The goals

adding C14 radiocarbon dates or archaeological/historical dates for host associated ancient metagenomes, update AMDir with open issues, the documentation and amdirt.

If you would like to join, please fill out the [form](#) by Monday 26th of January 2026.

SPAAM Blog Post Competition!



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SPAAM Blog Competition

Share your passion for ancient DNA in our first-ever blog post contest!

Theme: Anything related to ancient DNA

Length: 1000–2000 words

How to Enter: Email your post to spaam.community@gmail.com with the subject “blog post submission”

Deadline: 15 Feb 2026 11:59 PM CEST

There is a prize!



SUBMIT YOUR ENTRY NOW!



The SPAAM Community is thrilled to announce our first-ever blog post competition! We invite all members of the SPAAM community to submit an original blog post on a topic of their choice related to ancient DNA; whether it's a reflection on your research, an exciting methodological advance, an ethical discussion, or a creative take on the field.



What you need to know:

- Theme: Anything connected to ancient DNA
- Length: ~1000–2000 words, figures are welcome
- Audience: The broader ancient DNA community and interested readers
- Deadline: 15th February 2026
- Submission: please send your piece to the SPAAM email **spaam.community@gmail.com** with the title “blog post submission”



The winner:

You will not only get a chance of having your post featured on the SPAAM website, but also win a mystery book prize!



Publication:

All submissions will be considered for publication after an editing and proofreading process. This is a great opportunity to share your perspective, practice science communication, and engage with the community.



We can't wait to read your stories and insights. Let your creativity shine!

Introduction to Ancient Metagenomics 2025 edition

The 2025 edition of the [Ancient Metagenomics textbook](#) is now released!

This includes minor updates to some chapters, and one brand new shiny one that covers the content of the 'Introduction to NGS sequencing' section of the summer school! Reminder: this practical theory and bioinformatics textbook is open-access and you are welcome to use any/all of it in your own teaching!

Upcoming events

Conferences

EMBL Symposium: Reconstructing the human past: using ancient and modern genomics

- Location & Time: EMBL Heidelberg, Germany and Virtual 15-18 Sep 2026
- Abstract submission: 9 Jun 2026
- Registration (On-site): 4 Aug 2026
- Registration (Virtual): 8 Sep 2026

32nd Annual Meeting of the European Association of Archaeologists

- Location & Time: Athens, Greece 26-29 Aug 2026
- Abstract submission: 5 Feb 2026
- Registration: 29 Apr 2026

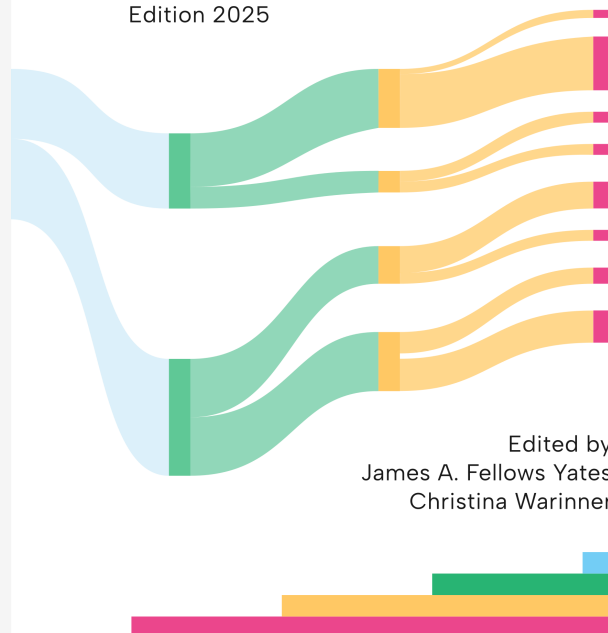
XXI World Congress of the International Union of Prehistoric and Protohistoric Sciences (UISPP)

- Location & Time: Poznań, Poland 31 Aug-4 Sep 2026
- Abstract submission: 9 Feb 2026
- Registration: 30 Apr 2026

Relevant Physalia courses

Introduction to Ancient Microbial Metagenomics

Edition 2025



Edited by
James A. Fellows Yates
Christina Warinner



[Comparative Genomics](#)

23-27 February 2026

This course offers a practical introduction to comparative genomics for both biologists and bioinformaticians. Participants will work with a variety of tools and workflows spanning genome assembly and annotation, detection of SNVs and structural variants, and evaluation of their functional and evolutionary significance.

[Introduction to population genomics](#)

16-20 March 2026

This five-day course provides an in-depth introduction to population genomics, combining conceptual teaching with hands-on analytical practice. Participants will explore a broad range of topics, from foundational bioinformatics and population structure to introgression, demographic inference, genome scans, and landscape genomics, beginning with core skills for managing large short-read datasets.

[AI for Genomics: From CNNs and LSTMs to Transformers](#)

7-9 April 2026

This course introduces participants to the use of contemporary AI methods for analysing genomic and metagenomic data. Through interactive coding sessions and guided notebooks, learners will practice building models for sequence analysis, identifying biologically relevant patterns, and interpreting model outputs. The training highlights practical applications, including predicting functional elements, classifying sequences and their origins, and generating new biological sequences.

[Conservation Genomics](#)

ONLINE, 7-10 April 2026

This course provides an introduction to applying population genomic approaches within conservation biology. Participants will be guided through designing genomic studies, generating and processing raw sequencing data, and filtering SNPs to build a workable dataset.

[Ancient Metagenomics](#)

4-6 May 2026

And our favourite course! Ancient Metagenomics introduces the major analytical challenges in ancient metagenomics and provides a practical, in-depth guide to implementing the aMeta workflow for robust ancient metagenomic analysis.

Open positions

PhD

- [University of East Anglia: Historical genomics of the declining red squirrel in Britain.](#) Deadline 7 January 2026
- [University of Aberdeen QUARTILES DLA: Oak genetics through time: ancient DNA analysis of sub-fossil oak for woodland management and conservation.](#) Deadline 16 January 2026
- [The University of Reading CR2026_39: Reconstructing the trophic ecology of Caribbean mammals: new baselines to understand island evolution, extinction and conservation.](#) Deadline 12 January 2026
- [University of Oxford Paleovirology: inferring ancient transmissions and consequences of ancient viral encounters.](#)

Deadline 8 January 2026

- [University of Ferrara: population genomics and metagenomics.](#)

Postdoc

- [University of Texas at Austin: e population genetics and natural selection of Central Eurasia.](#) Deadline 31 Dec 2025
- [Postdoctoral researcher in sedimentary DNA analysis of human-wildlife relationships.](#) Deadline 12 January 2026
- [MPI-EVA: Ancient Metagenomics and Microbiome Sciences.](#) Deadline 20 January 2026

Tenure-track

- [Assistant or Associate Professor in Genomics, Epigenetics, and Biomes at the University of Georgia, USA](#)
 - [Assistant Professor of Anthropology specializing in Bioarcheology at Utah State University, USA](#)
 - [Assistant/Associate Professor of Molecular Anthropology at the University of Oklahoma, USA](#)
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Funding opportunities

PhD Studentships

- [EMBL International PhD Programme](#) The next call for application will open by end of January 2026
- [PhD Studentships at the University of Reading, UK](#)

Postdoctoral Fellowships & Grants

- [EMBO Postdoctoral Fellowships](#): applications accepted all year-round, international mobility required. Next cut off date: 23 January 2026.
- [Wellcome Early-Career Awards](#): deadline - 17 February 2026
- [Leverhulme Early Career Fellowship \(UK\)](#): next round will open on January 1st 2026
- [UK Research and Innovation Future Leaders Fellowship - transition to independence](#): next round will open in February 2026 with a deadline in June 2026.

Small Grants

- [EMBO Scientific Exchange Grant](#): on a rolling basis, for 1-3 months of exchanges in EMBC countries
 - [Leakey Foundation research grants for work on human origins and evolution](#): deadline 10 January 2026
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Recent publications

- [Barbieri et al., Paratyphoid fever and relapsing fever in 1812 Napoleon's devastated army](#)
- [Blevins et al., Ancient DNA insights into diverse pathogens and their hosts](#)
- [Bolognesi et al., Optimizing ancient DNA recovery from archaeological plant seeds](#)
- [Capone et al., Targeted pathogen profiling of ancient feces reveals common enteric infections in the Rio Zape Valley, 725-920 CE](#)

- [Castellano-Hinojosa et al., Temporal succession of bacterial and archaeal communities in a Mediterranean high-mountain lake over the last 430 years using sedimentary DNA](#)
- [Chen et al., Fifteenth century CE Bolivian maize reveals genetic affinities with ancient Peruvian maize](#)
- [De Martino et al., The dispersal of domestic cats from North Africa to Europe around 2000 years ago](#)
- [De Sanctis et al., Bamdam: a post-mapping authentication toolkit for ancient metagenomics](#)
- [Gilardet et al., Paleogenomics Reveals a Loss of Bovine Lineages in Mid-latitude Asia Over the Last 200,000 Years](#)
- [Gwee et al., Performance of Two Custom Probe Kits for In-Solution Enrichment of Ancient Avian DNA](#)
- [Han et al., The late arrival of domestic cats in China via the Silk Road after 3,500 years of human-leopard cat commensalism](#)
- [Kessler et al., Ancient DNA of the Toronto Subway Deer adds to the extinction list of ice age megafauna](#)
- [Kocher et al., Insights into infectious diseases through ancient pathogen genomics](#)
- [Kraft et al., CarpeDeam: a de novo metagenome assembler for heavily damaged ancient datasets](#)
- [Ledger et al., Parasite infections at the Roman Fort of Vindolanda by Hadrian's Wall, UK](#)
- [Liu et al., Development and evaluation of an optimized Ancient DNA extraction method for femoral diaphyses and simulated heat-treated teet](#)
- [Mármol-Sánchez et al., Ancient RNA expression profiles from the extinct woolly mammoth](#)
- [Murph et al., First confirmation of Mycobacterium tuberculosis complex from medieval Ireland by aDNA analysis - palaeopathological and microbial findings](#)
- [Myburgh et al., Detection of Clostridium sporogenes in a Roman-era cattle mass grave at Vilauba](#)
- [Niemann et al., ParaRef: a decontaminated reference database for parasite detection in ancient and modern metagenomic datasets](#)

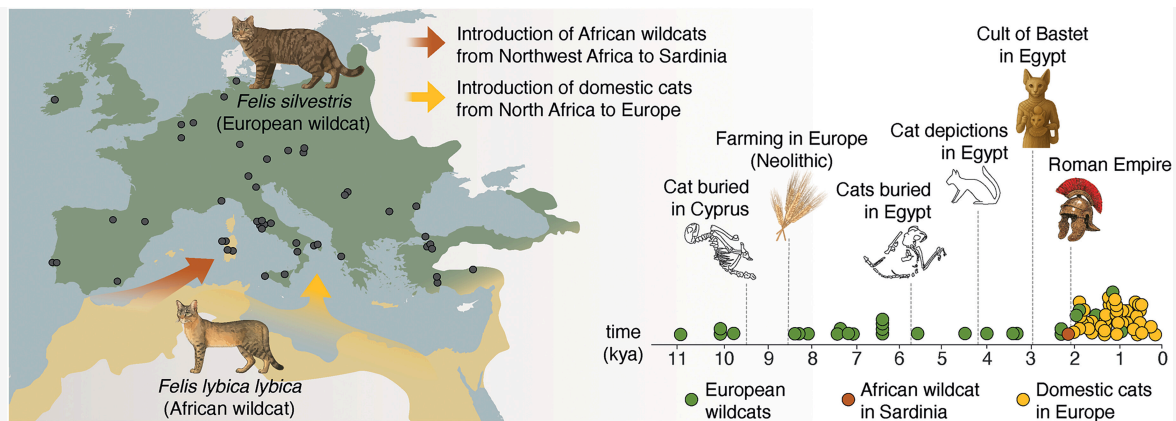
- [Oskolkov et al., Improving taxonomic inference from ancient environmental metagenomes by masking microbial-like regions in reference genomes](#)
 - [Przelomska et al., Population structure and inter-species admixture within a likely extinct yet formerly widespread Hawaiian honeycreeper](#)
 - [Sharko et al., Ancient DNA sheds light on the historical distribution of the rare and ephemeral plant *Coleanthus subtilis* in Southern Siberia](#)
 - [Szymanski et al., quick analysis of sedimentary ancient DNA using quicksand](#)
 - [Wright et al., Wet Lab Protocols Matter: Choice of DNA Extraction and Library Preparation Protocols Bias Ancient Oral Microbiome Recovery](#)
 - [Zhang et al., Revisiting the dispersal history of walnut by analyzing the ancient DNA of walnut remains in Xinjiang, China](#)
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Publication highlights

In this edition of Publication Highlights, we want to shed light on a fascinating article that was recently published in [Science](#)!

The paper by De Martino and colleagues took a closer look at how and when domestic cats conquered the world, telling us a time-travel story using ancient genomics. The authors generated and analysed 70 ancient cat genomes alongside 17 modern representatives. To anchor genetic patterns in time, 37 specimens were directly radiocarbon dated, providing a well-resolved genomic time transect covering ~11,000 years.

The dispersal of domestic cats from North Africa to Europe around 2000 years ago



What are the **key findings**?

- Cats carrying African wildcat mitochondrial lineages in Neolithic Europe were not early domestic cats, but European wildcats with ancient African admixture.
- The first clear genomic signal of cats belonging to the domestic cat gene pool appears around 2,000 years ago, coinciding with Classical Antiquity rather than the Neolithic.
- These early domestic cats show closest affinities to North African wildcats, supporting a rapid, human-mediated dispersal across the Mediterranean.
- North African wildcats were introduced to Sardinia during the first millennium BCE, earlier than previously thought.

By moving beyond mtDNA and tapping into genome-wide 🧬 data, this study gives us a much clearer picture of how cats 🐱 actually made their way into Europe. It nicely illustrates how deep-time admixture and hybridisation can muddy the waters when we rely on single genetic markers alone, and why palaeogenomics is such a powerful tool for untangling complex human–animal relationships that are otherwise hard to see in the archaeological record.

A *purr-fect* reminder that when it comes to ancient DNA, curiosity really *didn't* kill the cat! 🐱

Wow you made it all the way to the end!

Great job – we hope this email has something useful for you!

Until next time!

